



Phil1009

Logic and Critical Thinking

College of Social Sciences and Humanities
Department of Social Sciences

Course Outline

1st Semester 2025/2026

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1 Course Information

Course Code	Phil.1009
Course Name	Logic and Critical Thinking
Year/Semester	Freshman / I
Academic Year	2025/2026
Pre-requisite(s)	None
Co-requisite(s)	None
Credit Hour / ECTS	3
Course schedule	To Be Determined (TBD)

2 Instructor Information

2.1 Course Coordinator / Instructor

Name	Mohammed Z.
E-mail	mohammed.zeinu@aastu.edu.et
Consultation hour	Monday, Tuesday, and Thursday (Afternoon)
Office (Block/Room)	Block 80, Room 05
Telegram	t.me/zeinulabid

3 Course Description

Logic and Critical Thinking is an inquiry that focuses on the examination of arguments as its primary subject of investigation. This course introduces students to the fundamental concepts of logic and the methods of logical reasoning. Its primary objective is to equip students with essential skills for analyzing, evaluating, and constructing arguments. By honing these skills, students will enhance their ability to apply critical thinking in their thinking processes and written expression. By actively engaging with the material, students will cultivate intellectual curiosity and refine their ability to think critically, empowering them with invaluable skills that will serve them well in their academic endeavors and beyond.

4 Course Objectives

By the conclusion of the course, students will have achieved the following learning outcomes:

1. **Comprehend the interconnection between logic and philosophy:** Students will grasp the intricate relationship between logic and the broader field of philosophy and understand how logic serves as a foundational tool for philosophical inquiry.
2. **Identify key branches of philosophy:** Students will become familiar with the central areas of philosophy, including metaphysics, epistemology, ethics, and aesthetics, and recognize their significance.
3. **Recognize the importance of studying logic and philosophy:** Students will gain an appreciation for how these disciplines contribute to critical thinking, intellectual growth, and a deeper understanding of complex concepts.

4. **Understand fundamental logical concepts and arguments:** Students will acquire a solid understanding of basic logical concepts, such as premises, conclusions, validity, and soundness, and be able to distinguish between deductive and inductive reasoning.
5. **Identify fallacies in communication:** Students will develop the ability to identify common errors in reasoning and effectively critique flawed arguments in spoken and written communication.
6. **Cultivate linguistic precision:** Students will develop a heightened sensitivity to the importance of clear and accurate language use, refining their skills in constructing well-formed arguments.
7. **Construct sound arguments and evaluate arguments:** Students will develop the necessary skills to construct sound arguments and evaluate the strength and cogency of arguments presented by others.

5 Course Learning Outcomes (CLO) and Program Alignment

Upon successful completion of this course, students will be able to:

CLO 1	Define key philosophical and logical concepts, and differentiate between deductive and inductive reasoning.
CLO 2	Analyze arguments to identify logical fallacies and weaknesses in reasoning.
CLO 3	Construct sound, well-supported arguments and critique the arguments of others using logical principles.
CLO 4	Apply principles of critical thinking and linguistic precision to evaluate complex information and societal issues.

Program Learning Outcomes (PLO)

This course supports the development of the following graduate attributes:

PLO1 Engineering Knowledge	PLO7 Environment and sustainability
PLO2 Problem Analysis	PLO8 Ethics
PLO3 Design/development of solutions	PLO9 Individual and teamwork
PLO4 Investigation	PLO10 Communication
PLO5 Modern tool usage	PLO11 Project management and finance
PLO6 The engineer and society	PLO12 Lifelong learning

CLO-PLO Mapping Matrix

The following matrix shows the alignment between the Course Learning Outcomes and the Program Learning Outcomes.

CLO	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
CLO 1	✓											✓
CLO 2		✓		✓				✓				
CLO 3		✓	✓							✓		
CLO 4				✓		✓		✓		✓		✓

6 Course Schedule and Instructional Method

The course will be delivered through various modes to ensure an engaging and comprehensive learning experience, including lectures, group discussions, peer-learning activities, self-reading, and debates. The following schedule outlines the topics, activities, and learning outcomes for each week.

Week	Topic [Chapter]	Learning and Teaching Activity	CLO As-sessed
Week 1-2	Ch 1: Logic and Philosophy Meaning of Philosophy, Core Branches, Importance of Logic.	Lecture, Class Discussion, Self-Reading	CLO 1
Week 3-4	Ch 2: Basic Concepts of Logic Arguments, Premises, Conclusions, Deductive vs. Inductive reasoning, Evaluation criteria.	Lecture, Argument Analysis Exercises, Group Work	CLO 1, 3
Week 5	Ch 3: Logic and Language Meaning of terms, Types of definitions, Criteria for good definitions.	Lecture, Group Discussion on linguistic precision.	CLO 4
Week 6-7	Ch 4: Basic Concepts of Critical Thinking Meaning, Principles, Standards of good arguments, Factors affecting critical thinking.	Lecture, Case Studies, Peer Review of short arguments.	CLO 3, 4
Week 8	Mid-Term Examination	Mid-Exam	CLO 1, 4
Week 9-11	Ch 5: Logical Reasoning and Fallacies Formal vs. Informal fallacies, Fallacies of Relevance, Weak Induction, Presumption, and Ambiguity.	Lecture, Group Work ("Fallacy Hunting"), Student Presentations	CLO 2
Week 12-13	Ch 6: Categorical Propositions Components, Attributes, Venn Diagrams, Square of Oppositions, Logical Operations.	Lecture, Venn Diagram Practice, Problem Solving Sessions	CLO 3
Week 14-16	Course Review & Final Exam Preparation	Q&A Session, Review of key concepts and problem types.	All

7 Course Contents

Chapter One: Logic and Philosophy

- Meaning and Definition of Philosophy
- Core Branches of Philosophy
- Importance of Learning Logic and Philosophy

Chapter Two: Basic Concepts of Logic

- Basic Concepts of Logic (Premises, Conclusions, etc.)
- Techniques of Recognizing Arguments
- Types of Arguments: Deductive and Inductive
- Evaluation of Arguments: Validity, Soundness, Strength, Cogency

Chapter Three: Logic and Language

- Cognitive and Emotive Meaning of Words
- Intensional and Extensional Meaning of Terms
- Types, Purposes, and Techniques of Definition
- Criteria for Lexical Definitions

Chapter Four: Basic Concepts of Critical Thinking

- Meaning and Definition of Critical Thinking
- Principles and Standards of Good Argumentation
- Factors Affecting Critical Thinking
- Relevance of Critical Thinking

Chapter Five: Logical Reasoning and Fallacies

- Formal and Informal Fallacies
- Categories of Informal Fallacies: Relevance, Weak Induction, Presumption, Ambiguity, Grammatical Analogy

Chapter Six: Categorical Propositions

- Components and Attributes (Quality, Quantity, Distribution)
- Venn Diagrams and the Square of Oppositions
- Evaluating Immediate Inferences
- Logical Operations: Conversion, Obversion, and Contraposition

8 Assessment Methodologies

No	Assessment Method	Weight
1	Online Tests	25%
2	Paper Test	10%
3	Assignment & Presentation	10%
4	Attendance & Participation	5%
5	Final Exam	50%
Total		100%

9 Grading Policy

Grading will be according to the university's regulations. The letter grading system is shown below.

Marks	Grade	Grade Points
[90,100]	A+	4.00
[85,90)	A	4.00
[80,85)	A-	3.75
[75,80)	B+	3.50
[70,75)	B	3.00
[65,70)	B-	2.75
[60,65)	C+	2.50
[50,60)	C	2.00
<50	F	0.00

Mandatory Pass Requirement for Final Exam

Note: In accordance with the university senate decision (SEM 06/2015, September 8, 2023), it is mandatory for students to achieve a final exam result of **40% or higher** in order to receive a passing grade for the course. Failure to meet this requirement will result in a recorded grade of 'F,' regardless of the total mark out of 100%.

10 Course Policies

10.1 Attendance

Students are expected to attend every class. Absences hinder your progress and prevent participation. Attendance is marked as follows:

- **5/5:** Arrives on time and actively participates.
- **4/5:** Arrives on time but does not participate.
- **3/5:** Arrives late but participates, OR provides a valid excuse for absence.
- **2/5:** Arrives late and does not participate.
- **0/5:** Absent without a valid excuse.

10.2 Communications

Official channels for this course are e-mail, Canvas messaging, and in-class announcements. It is your responsibility to check these regularly. When emailing me or any professor, use a formal tone appropriate for a professional environment.

10.3 Assignment Submission and Formatting

Due Dates: All assignments must be submitted on the day they are due.

Late Submissions: Late submissions will not be accepted. An extension may be considered only if the student provides reasonable and tangible evidence for failing to submit on time.

Format: All submitted material must be typed or electronically composed (e.g., MS Office, Google Docs) and submitted as a PDF. Handwritten work will not be accepted. Your work must be proofread for spelling, grammar, and clarity.

10.4 Class Discussion and Participation

Active student participation is essential for this course. I am committed to creating a classroom environment where all students feel completely free to express their thoughts, ask questions, and challenge ideas without fear of sounding unsophisticated or "stupid." The best learning happens in a relaxed and open environment.

11 Academic Integrity

11.1 Academic Honesty

I will uphold and vigorously enforce the university's policies on academic honesty as described in the AASTU Senate Legislation of 2017. All work submitted must be your own. You are free to use reference material, but you must cite all sources properly. Copying any part of another's work and presenting it as your own is plagiarism. Any violation of this policy will result in immediate dismissal from the class with a failing grade. Remember: Allowing someone to copy your work is as dishonest as copying from someone else.

11.2 The Use of Artificial Intelligence (AI)

You are permitted to use generative AI tools to assist with your work, but you must adhere to two core principles: **Accountability** and **Transparency**.

- **Accountability:** You are 100% responsible for the work you submit. It is your job to fact-check, edit, and critically evaluate any AI-generated output. Submitting factually incorrect or poorly reasoned work is your failure, not the AI's.
- **Transparency:** You must cite any use of generative AI. At the end of any assignment where you used an AI tool, include an "AI Usage Statement" that clearly describes which tool you used, how you used it, and any key prompts you used. Failure to do so is a violation of the Academic Honesty policy.

12 Learning Resources

12.1 References

- [1.] Copi, I. M., Cohen, C., & McMahon, K. (2016). *Introduction to Logic* (14th ed.). Pearson.
- [2.] Hurley, P. J., & Watson, L. (2018). *A Concise Introduction to Logic* (13th ed.). Cengage Learning.
- [3.] Browne, M. N., & Keeley, S. M. (2018). *Asking the Right Questions: A Guide to Critical Thinking* (12th ed.). Pearson.
- [4.] Damer, T. E. (2008). *Attacking Faulty Reasoning: A Practical Guide to Fallacy-Free Arguments* (6th ed.). Wadsworth Publishing.
- [5.] Weston, A. (2018). *A Rulebook for Arguments* (5th ed.). Hackett Publishing Company.

12.2 Online Resources

- [1.] Stanford Encyclopedia of Philosophy: A comprehensive and authoritative resource.
<https://plato.stanford.edu/>
- [2.] Internet Encyclopedia of Philosophy: Peer-reviewed articles on key topics.
<https://iep.utm.edu/>
- [3.] The Fallacy Files: An extensive guide to logical fallacies.
<https://www.fallacyfiles.org/>
- [4.] Purdue Online Writing Lab (OWL): Resources on argumentation and logic.
<https://owl.purdue.edu/>

13 Safety and Securities

Report all emergencies or suspicious activity to the office of the Social Science Department:

- **Location:** Block-80, Office number-117
- **Email:** [social sciences department](#)
- **Website:** Department of Social sciences - [Website](#)